It's Not (Only) About Optimization: Collaborative Learning as a Data Pipeline Problem

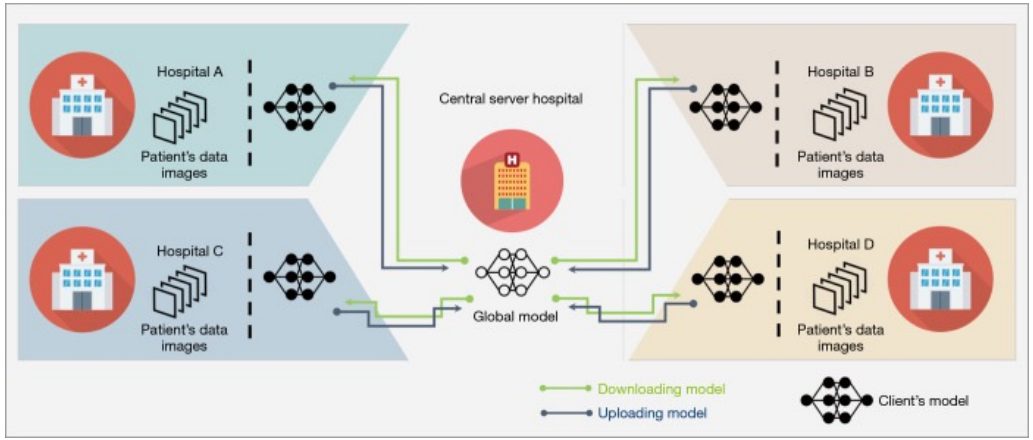
Lesson learnt from the Fed-BioMed open-source initiative

Marco Lorenzi

Inria Sophia Antipolis, Université Côte d'Azur

Epione Research Group

Federated learning for data governance



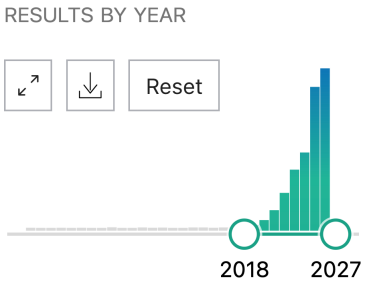
From Ng et al, Quant Imaging Med Surg, 2021;

From Xu et al. J Healthc Inform Res. 2020

Table 1

Summary of recent work on federated learning for healthcare

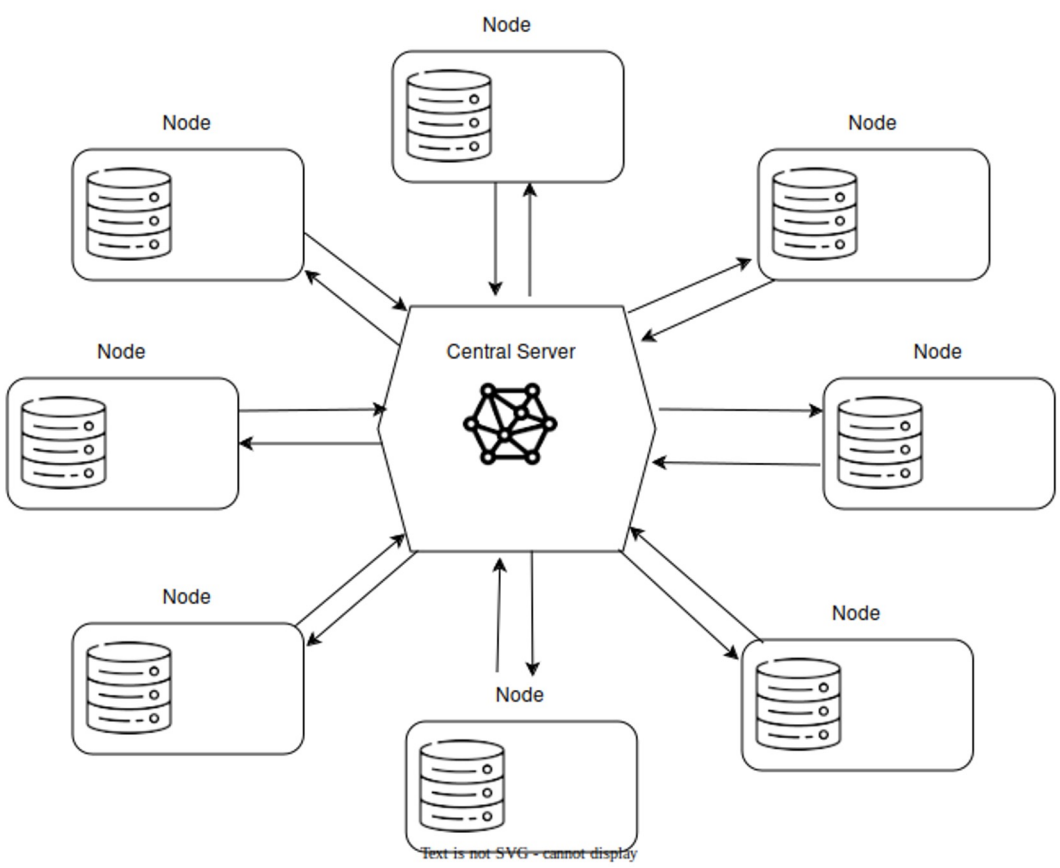
Problem	ML method	No. of clients	Data
Patient similarity learning [62]	Hashing	3	MIMIC-III [50]
Patient similarity learning [108]	Hashing	20	MIMIC-III
Phenotyping [55]	TF	1–5	MIMIC-III, UCSD [104]
Phenotyping [67]	NLP	10	MIMIC-III
Representation learning [93]	PCA	10–100	ADNI, UK Biobank, PPMI, MIRIAD
Mortality prediction [45]	Autoencoder	5–50	eICU Collaborative Research Database [81]
Hospitalization prediction [10]	SVM	5, 10	Boston Medical Center
Preterm-birth prediction [9]	RNN	50	Cerner Health Facts
Mortality prediction [80]	LR, NN	31	eICU Collaborative Research Database
Mortality prediction [90]	LR, MLP	2	MIMIC-III
Activity recognition [16]	CNN	5	UCI Smartphone [4]
Adverse drug reactions Prediction [19, 20]	SVM, MLP, LR	10	LCED, MIMIC
Arrhythmia detection [110]	NN	16, 32, 64	PhysioNet Dataset [21]
Disease prediction [33]	NN	5, 10	Pima Indians Diabetes Dataset [95], Cleveland Heart Disease Database [23]
Imaging data analysis	VAE	4	MNIST, Brain Imaging Data
Mortality prediction [101]	LRR, MLP, LASSO	5	Mount Sinai COVID-19 Dataset



PubMed query “federated learning”

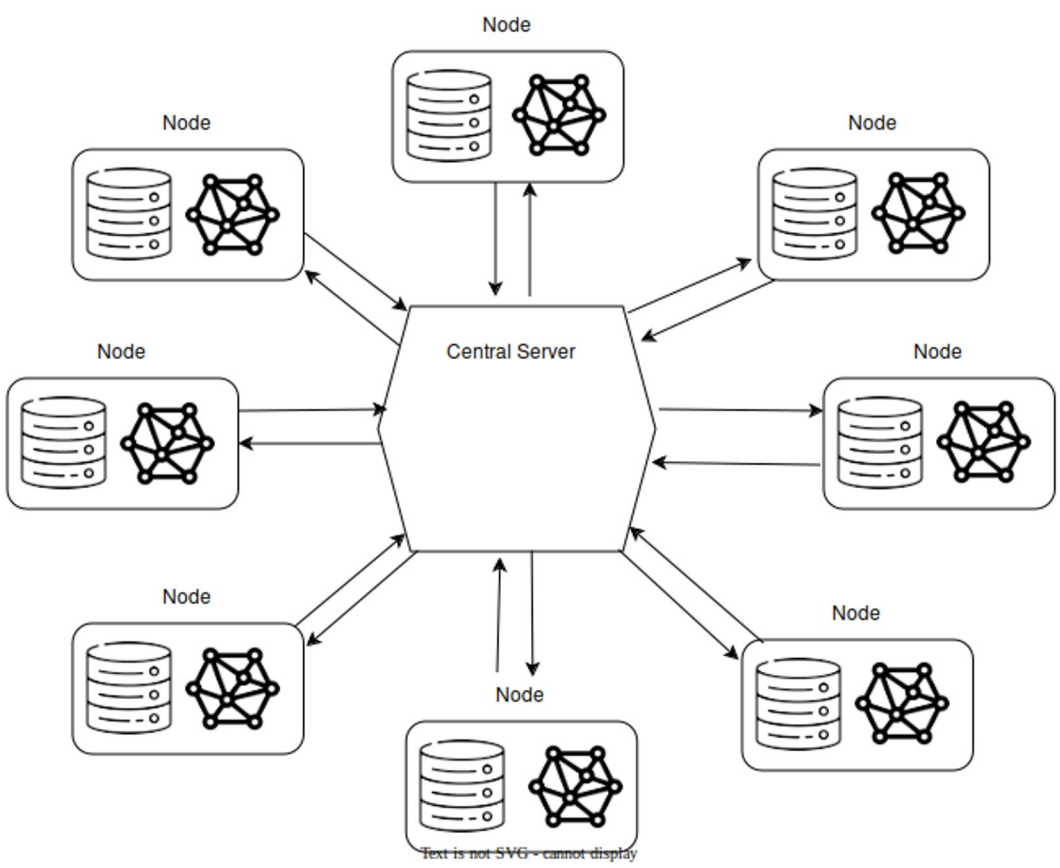
Rieke et al. NPJ Digit Med. 2020; Xu et al. J Healthc Inform Res. 2020

Step 1:
A global model is instantiate by the Central Server

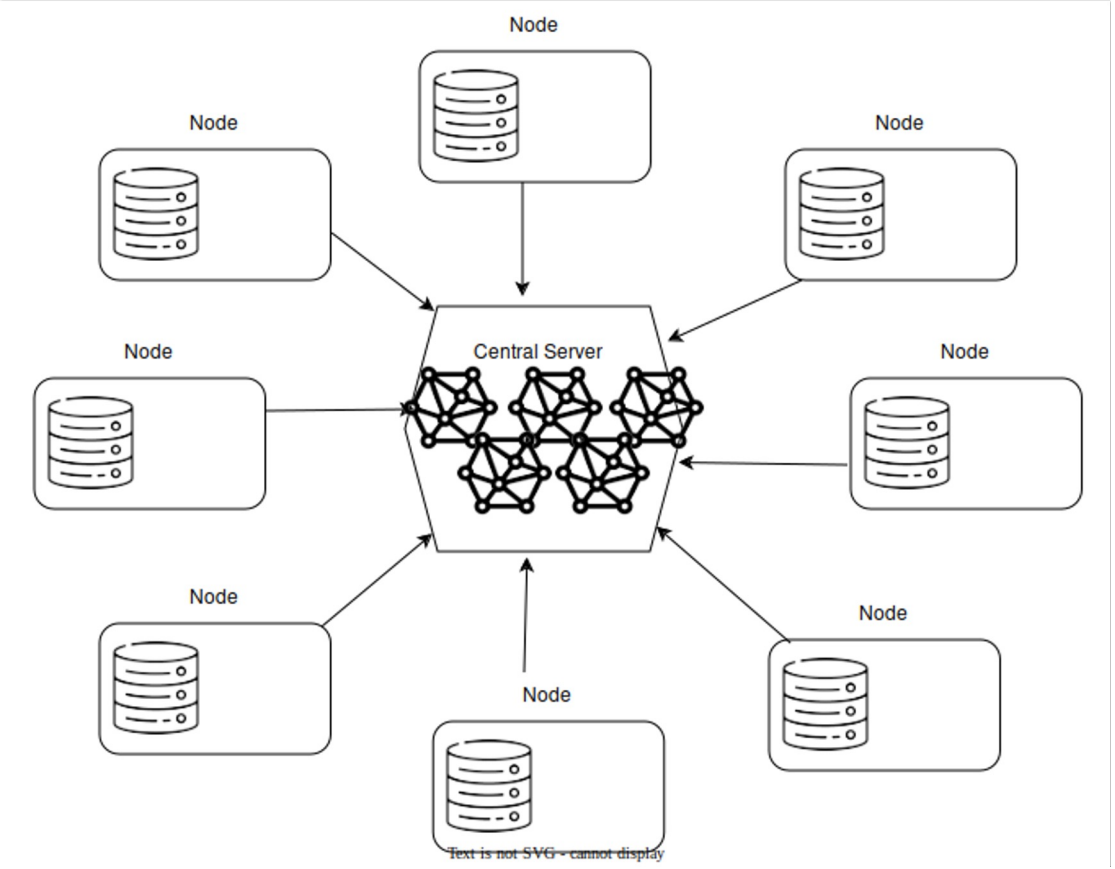


Step 2:
Global model is sent to all Nodes involved in the training.

Each Node receives a copy of the global model (local model) and will train their model w.r.t their local dataset

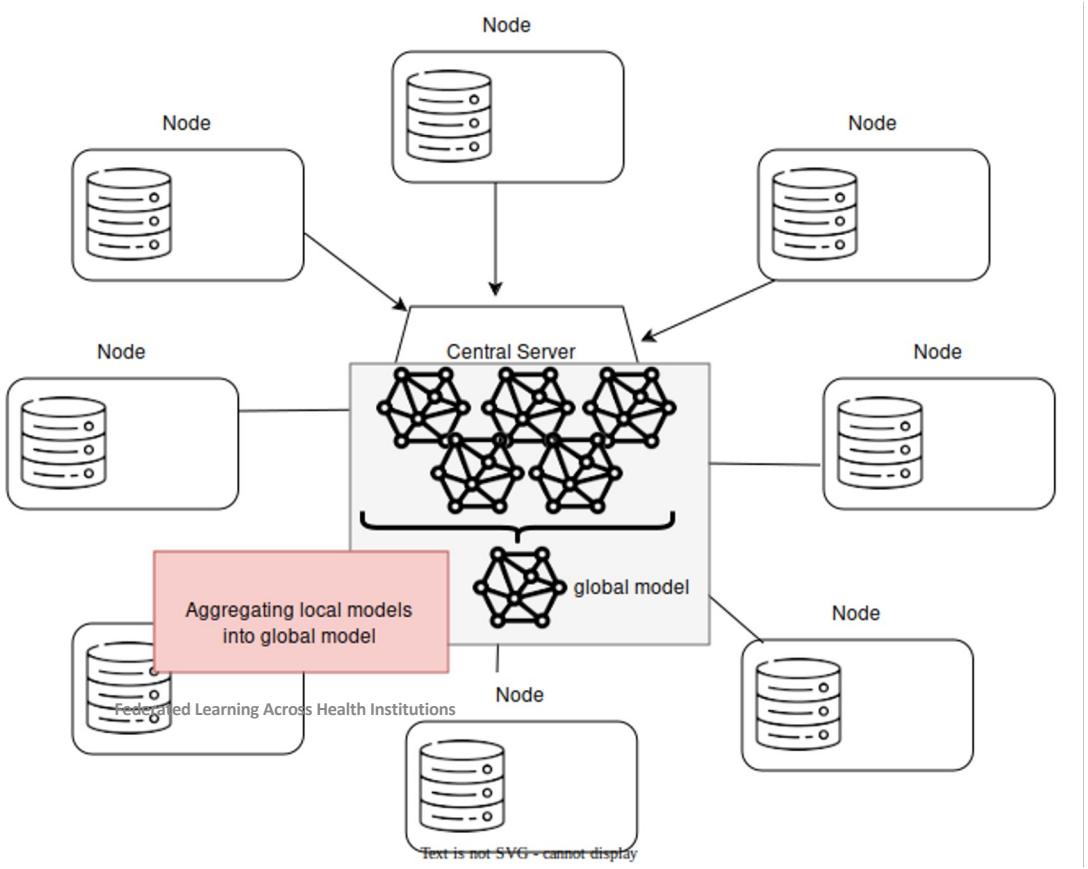


Step 3:
Local models are sent back to the Central Server

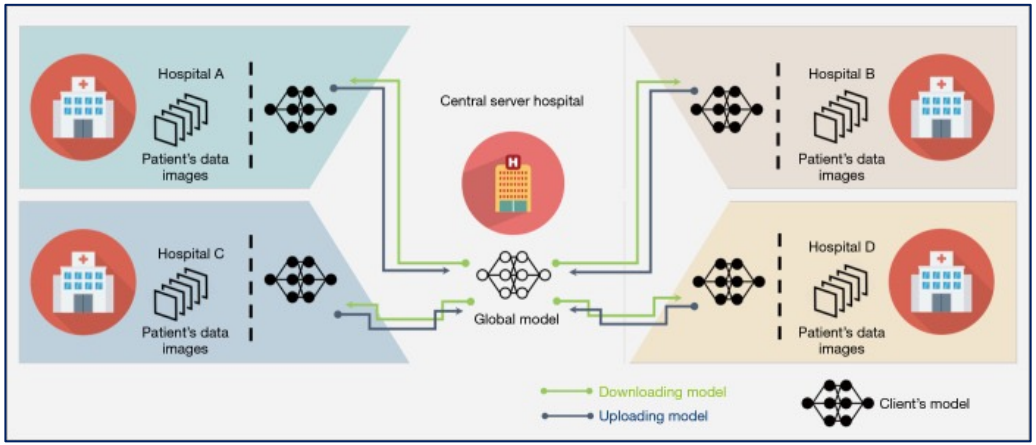


Step 4:
Local models are aggregated into a global model using an Aggregation method.

Steps 1 to 4 are repeated until convergence of the global model



Translation of Collaborative Medical Data Analysis

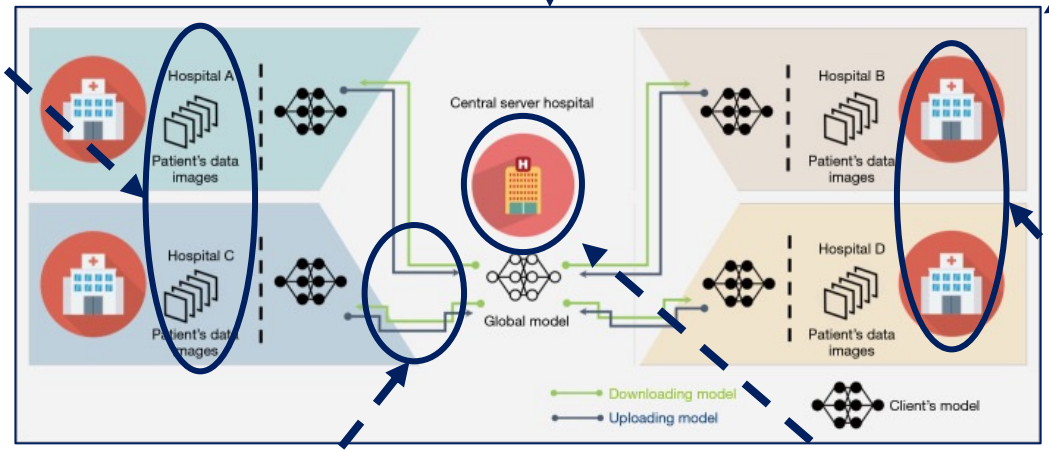


Translation of Collaborative Medical Data Analysis

Heterogeneous networks / database / data format

Security model

Regulation: Data Protection Officer, Homologation, Ethical Committee



Availability of local resources (hardware and personnel)

Data flow

Governance: Server location, Researcher role, Hospital control over FL

Fed-BioMed

Federated Learning for Healthcare by Design

Security

- Clients authentication
- Secured communications
- Model verification
- Differential Privacy
- Secure aggregation

Client control / Governance

- Experiment opt-in / -out
- Monitoring Tools
- Data verification/ pre-processing
- GUI
- Handling heterogeneous data types

Usability

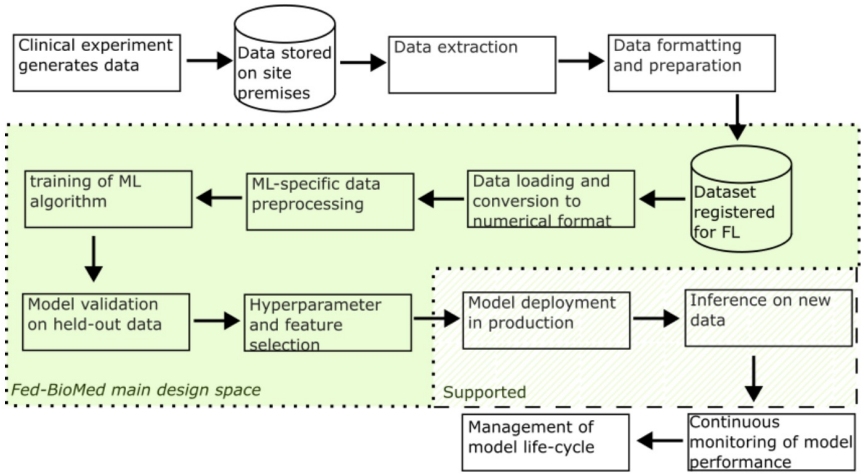
- Numpy/Pytorch/MONAI/sklearn compatible
- Easy control with Jupyter notebook
- Breakpoints and control of experiment
- Error handling
- FL aggregation and sampling strategies
- FL simulator

Community-driven

- Roadmap inspired by collaborating hospitals
- Long term planning and industrial support
- Commercial-friendly license (Apache 2.0)

Requirements & Design space

Challenges	Primary requirements	Secondary and minor
• Data privacy and protection • Siloed data • Heterogeneous, unstructured data • Integration of multiple data modalities • Dynamic model development cycle • Cyberattacks • Model and gradient attacks • Suboptimal IT infrastructure and computing capabilities • IP protection	**Data and model governance** Powerful and easy-to-use governance tools for nodes. **Integration with biomedical data sources** Support biomedical and interoperability standards; support diverse computing infrastructures. **Researcher interactivity** Support a highly interactive, dynamic model development cycle steered by the researcher. **Security** Secure software and related IT infrastructure against hacking; support model and gradient protection strategies.	federated pre/post processing support for several ML libraries portability drop-out tolerance high availability scalability lightweight



Cremonesi F., et al. *Federated Learning Systems, 832, 19-41, 2025*

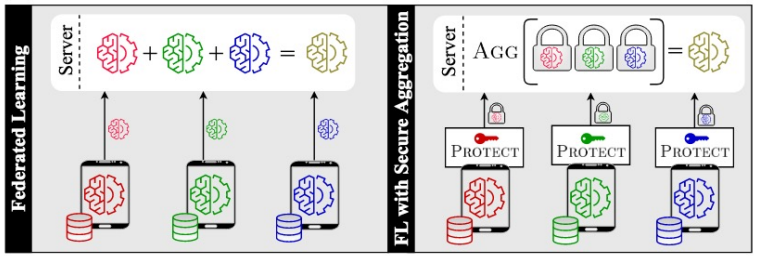
Community driven features

Community driven features

Secure Aggregation

Low Overhead Masking

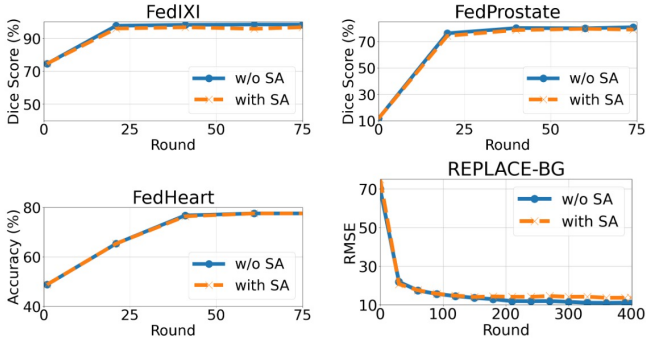
Ready to deploy



Generally < 1% overhead

SA	Time Train (s)	Time Enc. (s)	Time Tot. (s)
FedIXI ($d = 246K; n_{tot} = n = 3$)			
JL	68.10 ± 2.17	52.21 ± 0.85	121.48 ± 2.50
LOM	46.51 ± 1.39	0.62 ± 0.14	48.22 ± 1.03
FedHeart ($d = 258; n_{tot} = n = 4$)			
JL	0.24 ± 0.08	0.08 ± 0.01	0.68 ± 0.09
LOM	0.20 ± 0.09	> 0.01	0.59 ± 0.08
REPLACE-BG ($d = 256K; n_{tot} = 180; n = 18$)			
JL	N/A	N/A	N/A
LOM	53.72 ± 8.61	0.39 ± 0.06	57.42 ± 6.95
FedProstate ($d = 7.4M; n_{tot} = n = 4$)			
JL	N/A	> 300	> 300
LOM	7.65 ± 1.6	9.22 ± 0.38	23.86 ± 2.1

Loss accuracy < 2%



Taiello R., et al. *DeCaF MICCAI 2024, LNCS, 15274, 204-214.*

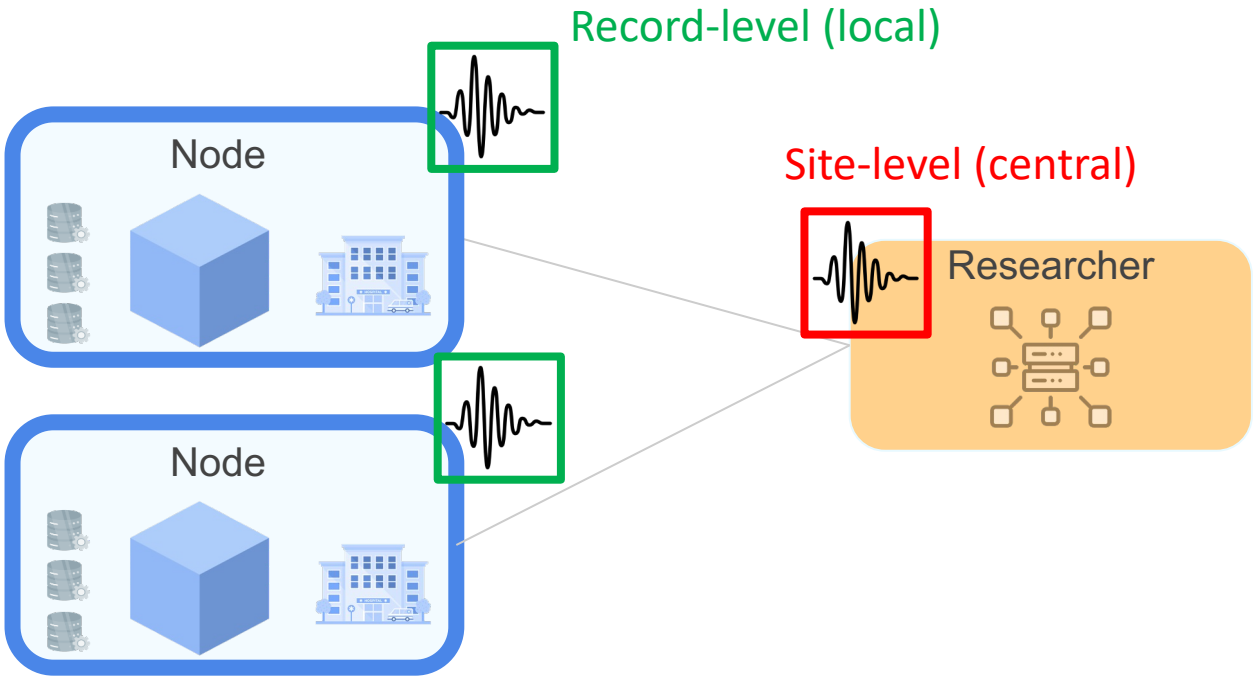
Community driven features

Secure Aggregation

Low Overhead Masking
Ready to deploy

Differential privacy

Local (record level)
Central (site level)



Community driven features

Secure Aggregation
Low Overhead Masking
Ready to deploy

Differential privacy
Local (record level)
Central (site level)

Experiment approval
Review interface
Decide what runs on the node

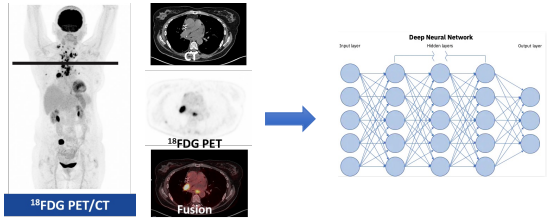


Communication security
VPN encapsulation
Outbound communications from hospitals only
Secure node to node communications

UNICANCER Consortium

18 FDG-PET Analysis for Predicting Treatment Response in Lung Cancer



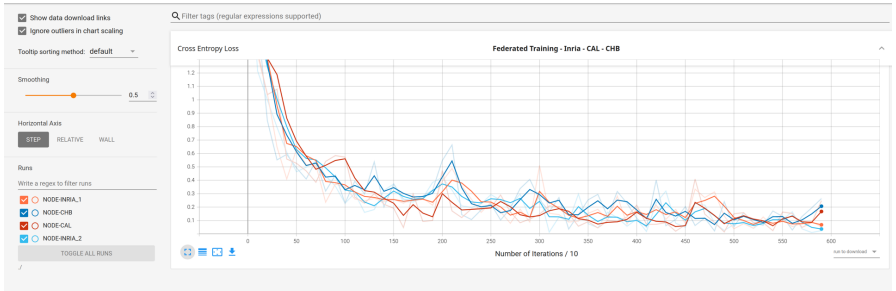


Humbert et al. Eur J Nucl Med Mol Imaging. 2020

Antoine Lacassagne
CENTRE DE LUTTE CONTRE LE CANCER
unicancer NICE

CENTRE de Lutte Contre le Cancer HENRI BECQUEREL
unicancer NORMANDIE - ROUEN

institutCurie



Prix Unicancer de l'Innovation 2024



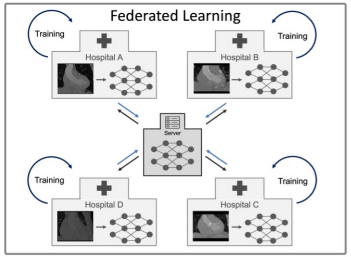
Article | Open access | Published: 06 February 2025

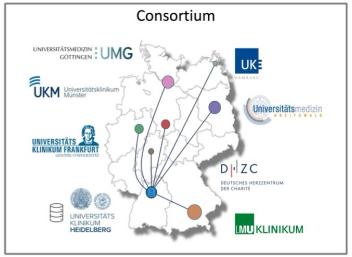
Real world federated learning with a knowledge distilled transformer for cardiac CT imaging

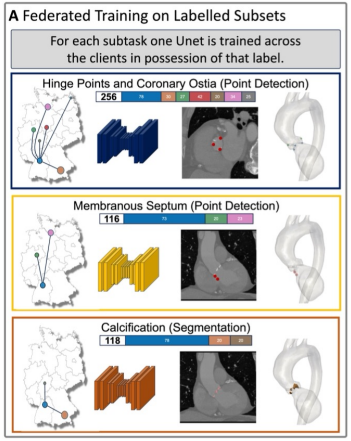
Malte Tölle ✉, Philipp Garthe, Clemens Scherer, Jan Moritz Seliger, Andreas Leha, Nina Krüger, Stefan Simm, Simon Martin, Sebastian Eble, Halvar Kelm, Moritz Bednorz, Florian André, Peter Bannas, Gerhard Diller, Norbert Frey, Stefan Groß, Anja Hennemuth, Lars Kaderali, Alexander Meyer, Eike Nagel, Stefan Orwat, Moritz Seiffert, Tim Friede, Tim Seidler & Sandy Engelhardt

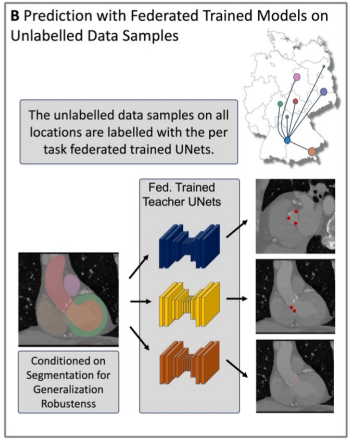
npj Digital Medicine **8**, Article number: 88 (2025) | Cite this article

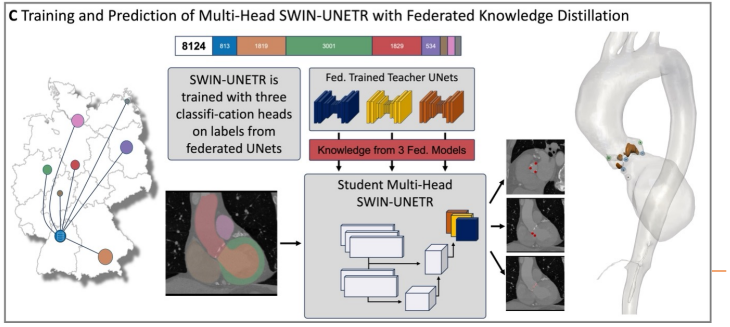
- Heidelberg University Hospital (server and client)
- University Medicine Göttingen
- LMU University Hospital Munich
- University Hospital Münster
- University Medical Center Hamburg-Eppendorf (UKE)
- Deutsches Herzzentrum der Charité (DHZC)
- University Medicine Greifswald
- University Medicine Frankfurt











DZHK
Deutsches Zentrum für Herz-Kreislauf-Forschung

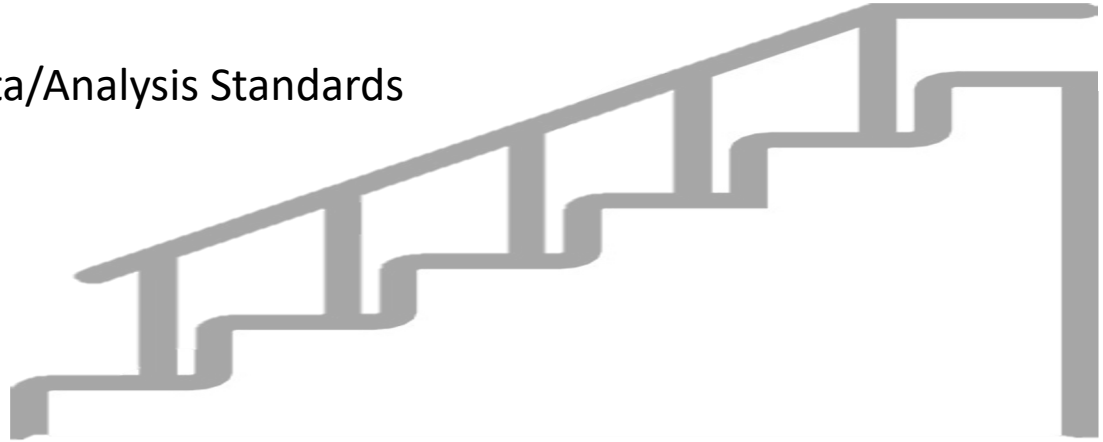
The collaborative learning pipeline

Cost effectiveness

Harmonization

Data/Analysis Standards

Data Access





The collaborative learning pipeline

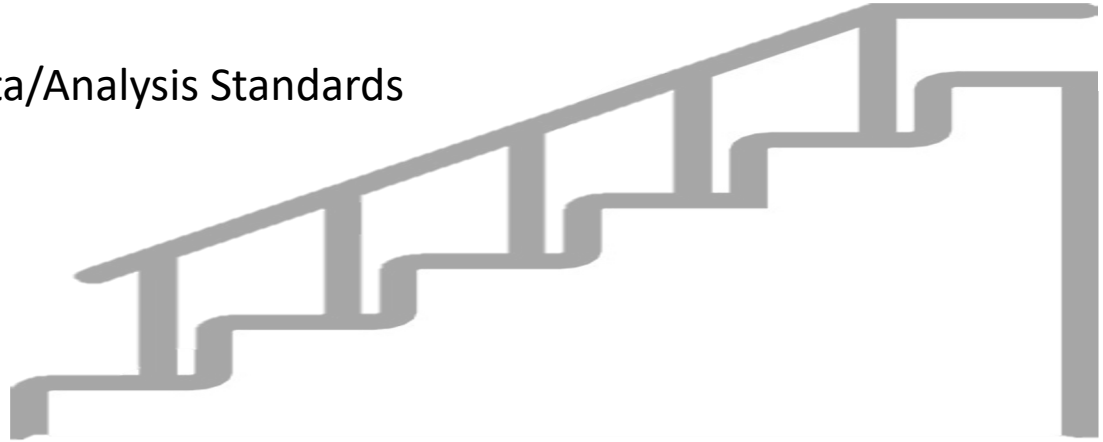
Cost effectiveness



Harmonization

Data/Analysis Standards

Data Access



Computer Science > Computers and Society

[Submitted on 7 Dec 2025 (v1), last revised 25 May 2026 (this version, v3)]

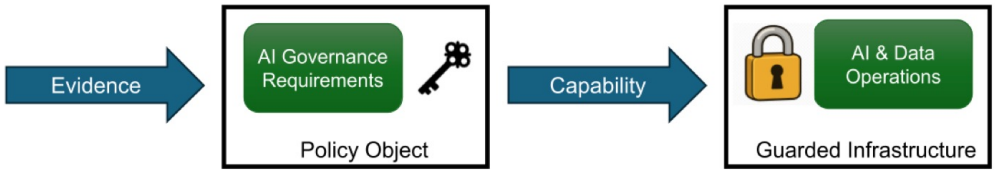
A Technical Policy Blueprint for Trustworthy Decentralized AI

Hasan Kassem, Orion Banks, Omar Benjelloun, Sergen Cansiz, Brandon Edwards, Patrick Foley, Inken Hagestedt, Taeho Jung, Peter Kairouz, Marco Lorenzi, Peter Mattson, Prakash Moorthy, Ann K Novakowski, Michael O'Connor, Bruno Rodrigues, Holger Roth, Micah Sheller, Dimitris Stripelis, Renato Umeton, Marc Vesin, Wenbin Zhang, Mic Bowman, Alexandros Karargyris

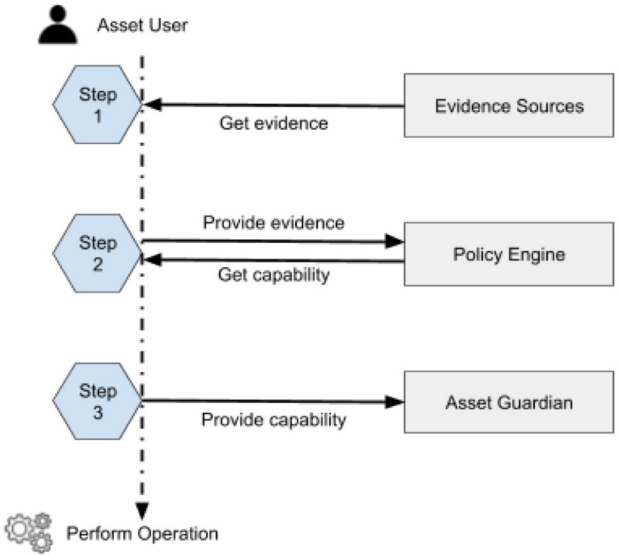
NeurIPS Dataset and Benchmarks Track 2026

ML Commons

Machine readable standard for policy rules



Automated policy workflow



Evidence Signatures, payment proofs, hardware attestations

Runs the policy object and verifies the evidence

Capability package: a signed permission for one operation

Applies the capability

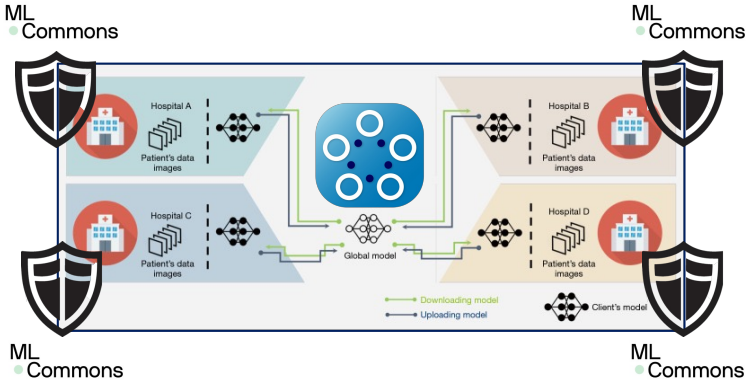
Shared policy objects make rules portable and auditable

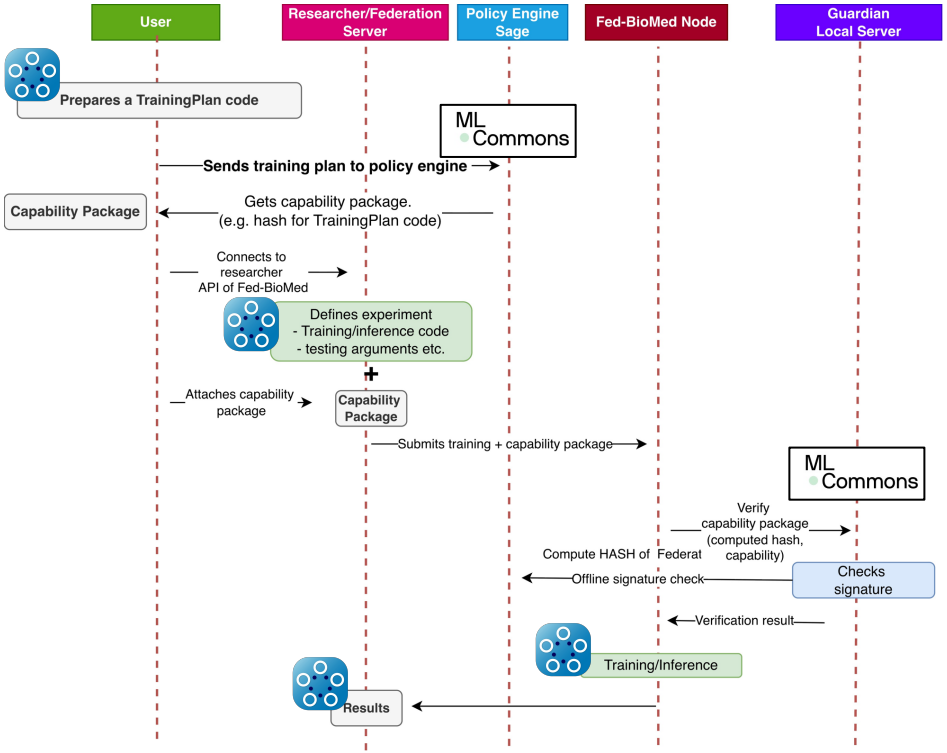
```
# illustrative hospital policy
allow: federated_averaging
deny: federated_sgd
require: other_providers >= 1
require: research_intent_verified
```

The data owner decides which optimizer may run.

Checks on the algorithm, hyperparameters and the number of sites can be automated.

Shared policy objects make rules portable and auditable





The collaborative learning pipeline

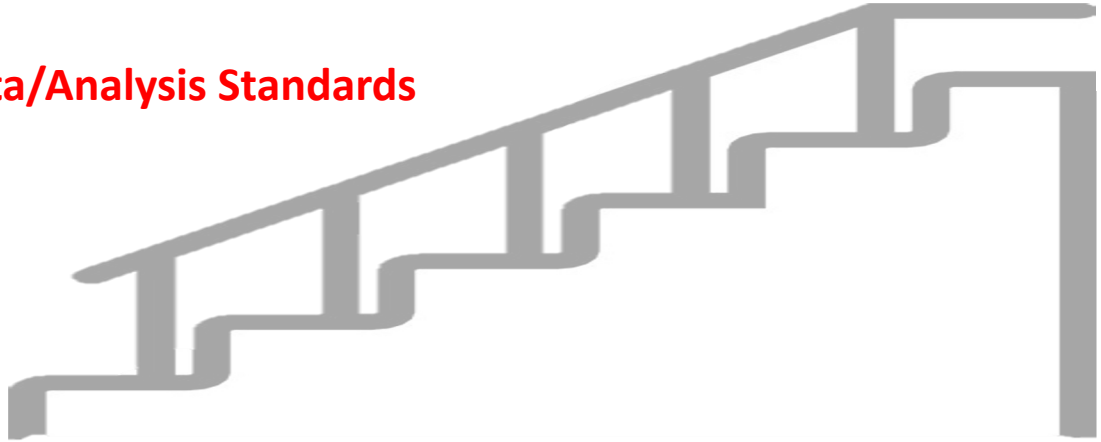
Cost effectiveness

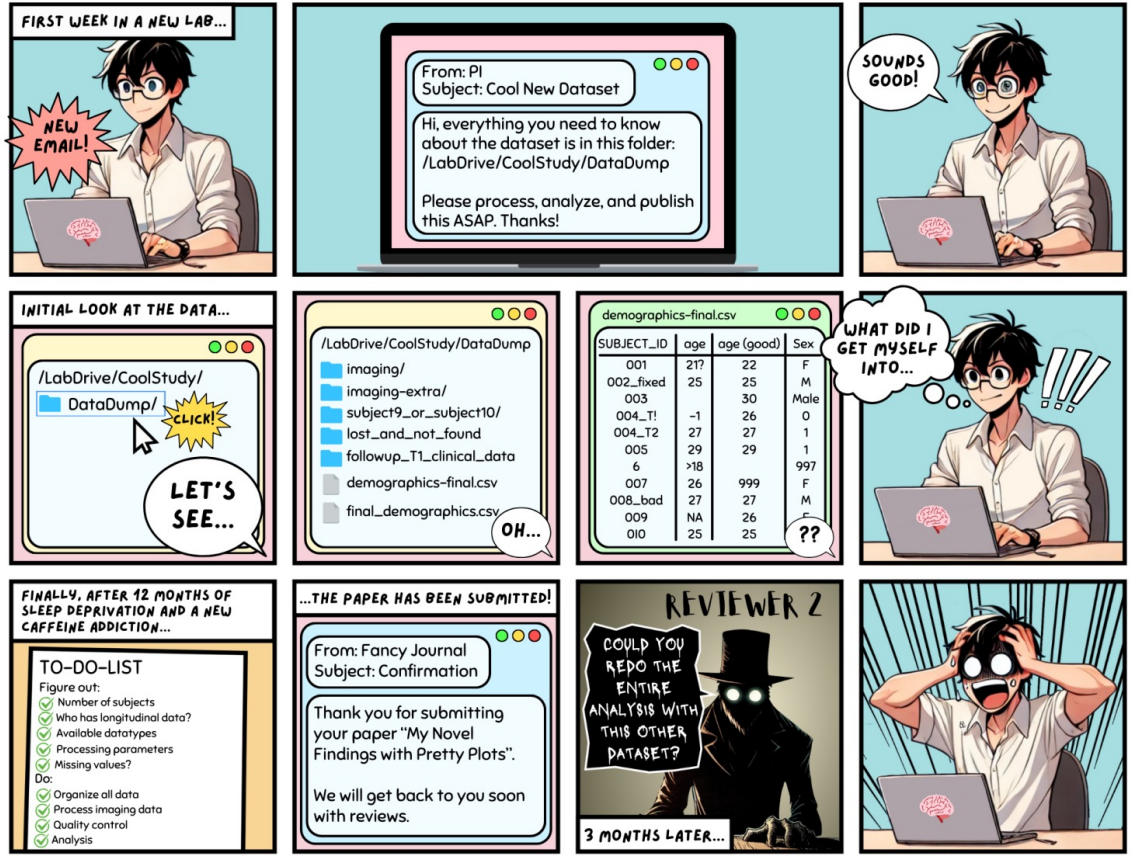


Harmonization

Data/Analysis Standards

Data Access





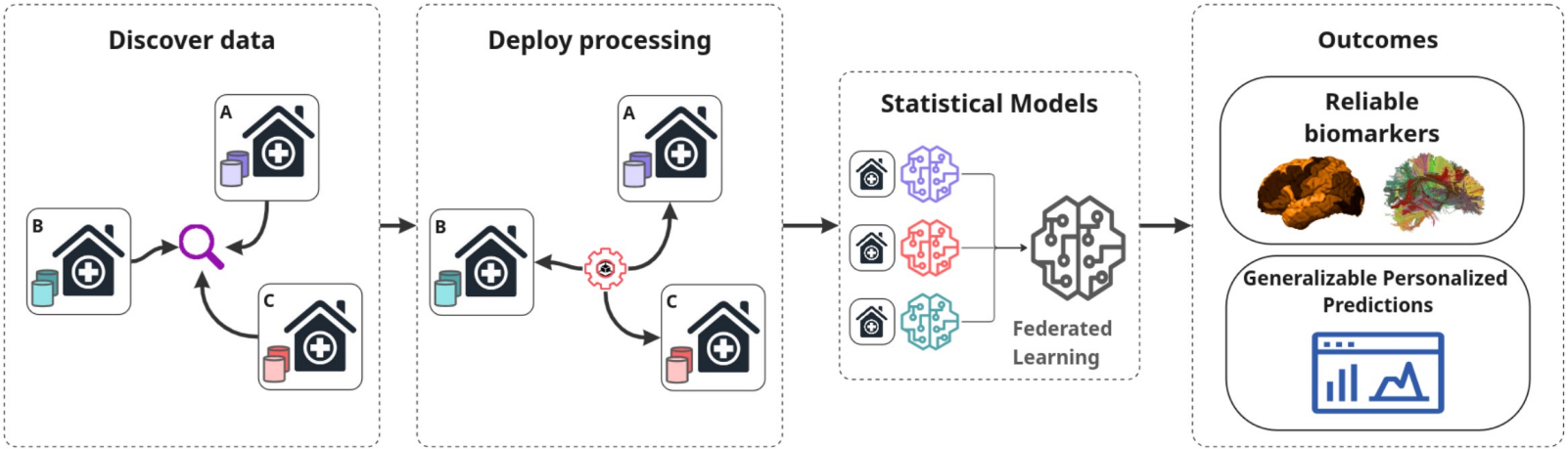
https://nipoppy.readthedocs.io/en/latest/index.html

Open neuroinformatics infrastructure ecosystem for federated multisite studies

bioRxiv
THE PREPRINT SERVER FOR BIOLOGY

Michelle Wang, Nikhil Bhagwat, Francesco Cremonesi, Mathieu Dugré, Julia-Katharina Pfarr, Emile d'Angremont, Alyssa Dai, Arman Jahanpour, Sebastian Urchs, Sergen Cansiz, Lucie Chambon, Ali Tolga Dinçer, Jhonatan Torres, Marc Vesin, Gabriel Pinilla-Monsalve, Yuan Song, Chris Vriend, Francis Jeanson, Oury Monchi, Ysbrand D. van der Werf, Marco Lorenzi, Jean-Baptiste Poline

doi: https://doi.org/10.64898/2026.04.30.721944



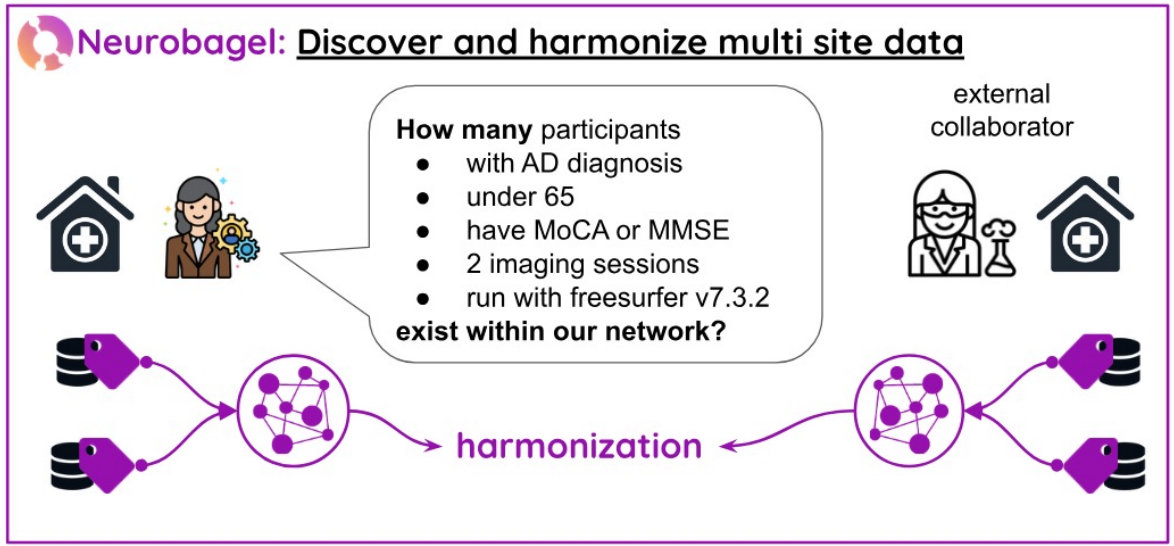
ORIGAMI

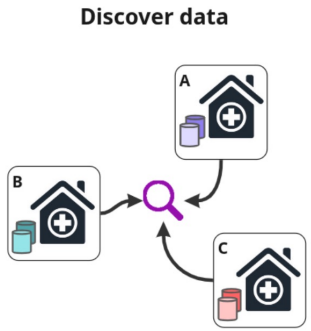
neuro
Montreal Neurological Institute-Hospital

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Discover and harmonize formats

https://neurobagel.org/





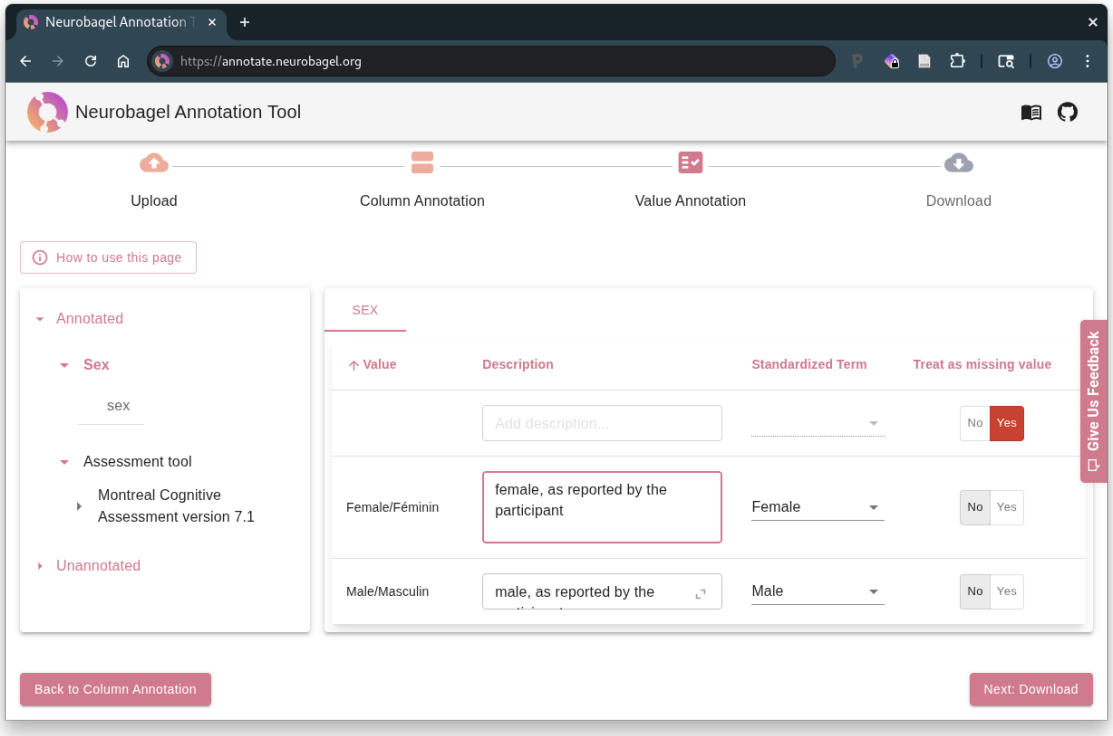
ORIGAMI

neuro
Montreal Neurological Institute-Hospital

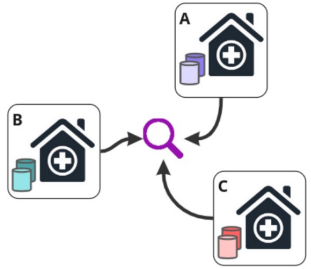
McGill UNIVERSITY

Discover and harmonize formats

https://neurobagel.org/



Discover data



sets harmonized vocabulary and semantics
supports federated discovery
enables cohort selection

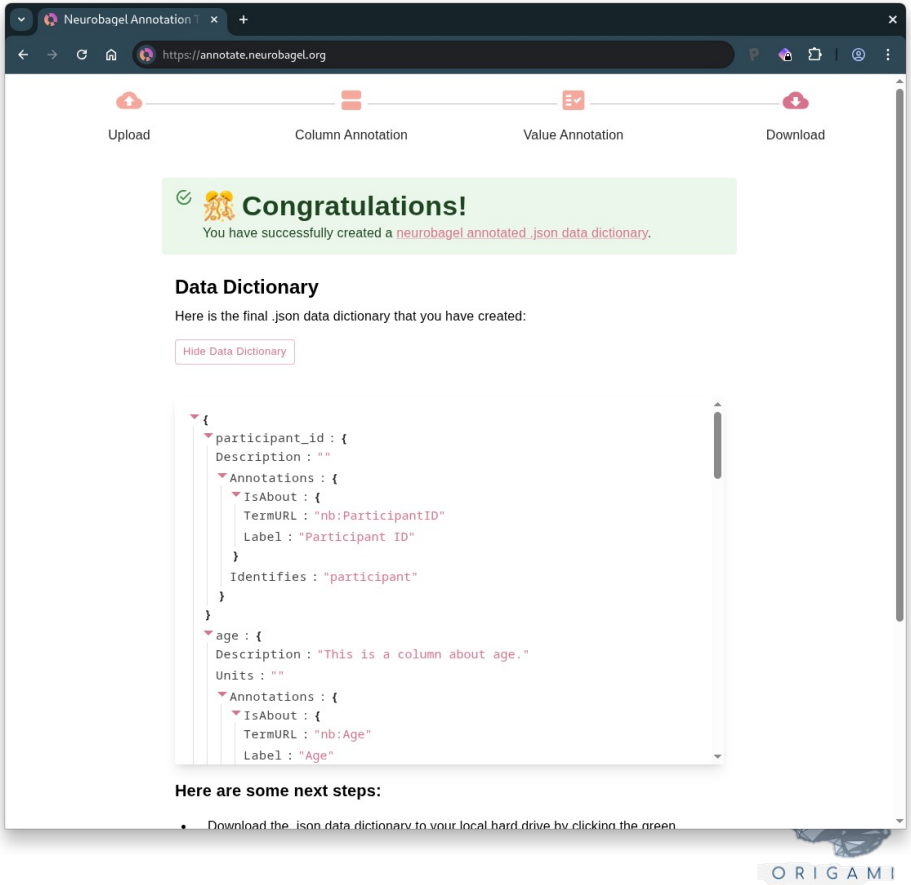
ORIGAMI

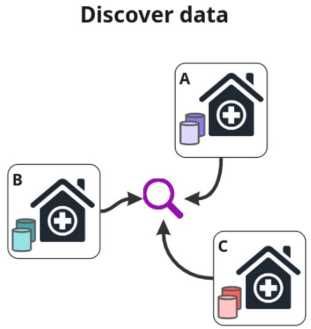
neuro
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Discover and harmonize formats

https://neurobagel.org/





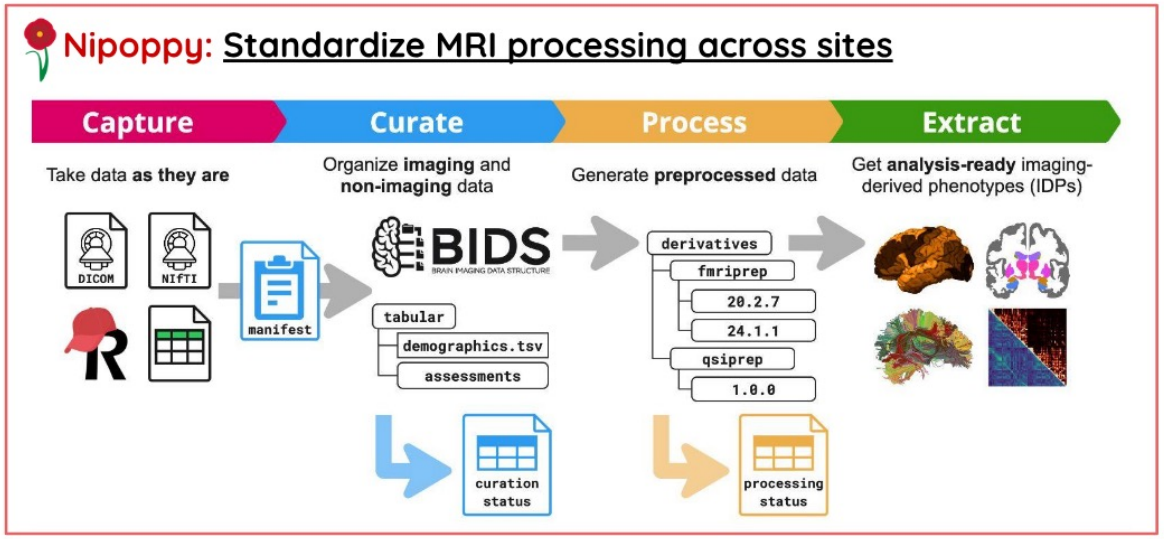
sets harmonized vocabulary and semantics
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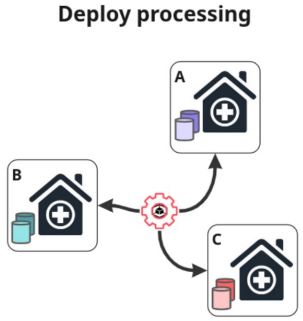
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Standardize pipelines

https://nipoppy.readthedocs.io/en/stable/





ORIGAMI

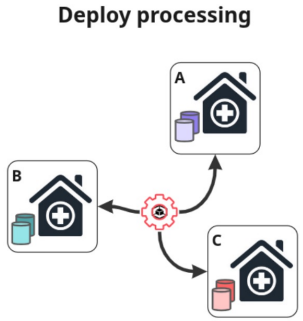
neuro
Montreal Neurological Institute-Hospital

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Standardize pipelines

https://nipoppy.readthedocs.io/en/stable/

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        },
        "BIND_PATHS": [
            "[[FREESURFER_LICENSE_FILE]]",
            "[[TEMPLATEFLOW_HOME]]",
            "[[NIPOPPY_DPATH_PIPELINE_WORK]]:/work",
            "[[NIPOPPY_DPATH_DERIVATIVES]]/freesurfer/7.3.2/output/[[NIPOPPY_BIDS_SESS
        ]
    },
    "STEPS": [
        {
            "INVOCATION_FILE": "invocation.json",
            "DESCRIPTOR_FILE": "descriptor.json",
            "HPC_CONFIG_FILE": "hpc.json",
            "TRACKER_CONFIG_FILE": "tracker.json"
        }
    ],
    "VARIABLES": {
        "FREESURFER_LICENSE_FILE": "Path to FreeSurfer license file",
        "TEMPLATEFLOW_HOME": "Path to the directory where TemplateFlow will store temp
    },
    "PIPELINE_TYPE": "processing",
    "SCHEMA_VERSION": "1.0"
}
```



inherits Neurobagel vocabulary and semantics
standardizes local data
harmonizes data preprocessing

ORIGAMI

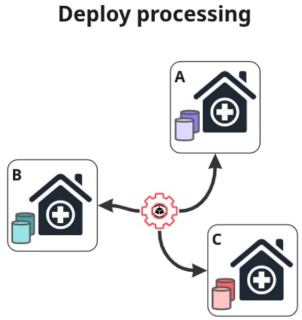
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Standardize pipelines

https://nipoppy.readthedocs.io/en/stable/

Zenodo ID	Title	Description
15299437	xcp-d postprocessing pipeline	Config and Tracker files for running xcp-d
15306681	freesurfer-7.3.2	Nipoppy configuration files for freesurfer 7.3.2 pipeline
15306657	dcm2bids-3.1.0	Nipoppy configuration files for dcm2bids 3.1.0 pipeline
15306667	static_FC-0.1.0	Nipoppy configuration files for static_FC 0.1.0 pipeline
15306677	fmriprep-24.1.1	Nipoppy configuration files for fmriprep 24.1.1 pipeline
15271392	Fmriprep Ciftify Pipeline	Config and Tracker files for running Ciftify
15306675	fmriprep-23.1.3	Nipoppy configuration files for fmriprep 23.1.3 pipeline
15306661	dcm2bids-3.2.0	Nipoppy configuration files for dcm2bids 3.2.0 pipeline
15306679	freesurfer-6.0.1	Nipoppy configuration files for freesurfer 6.0.1 pipeline
15306685	qsiprep-0.23.0	Nipoppy configuration files for qsiprep 0.23.0 pipeline



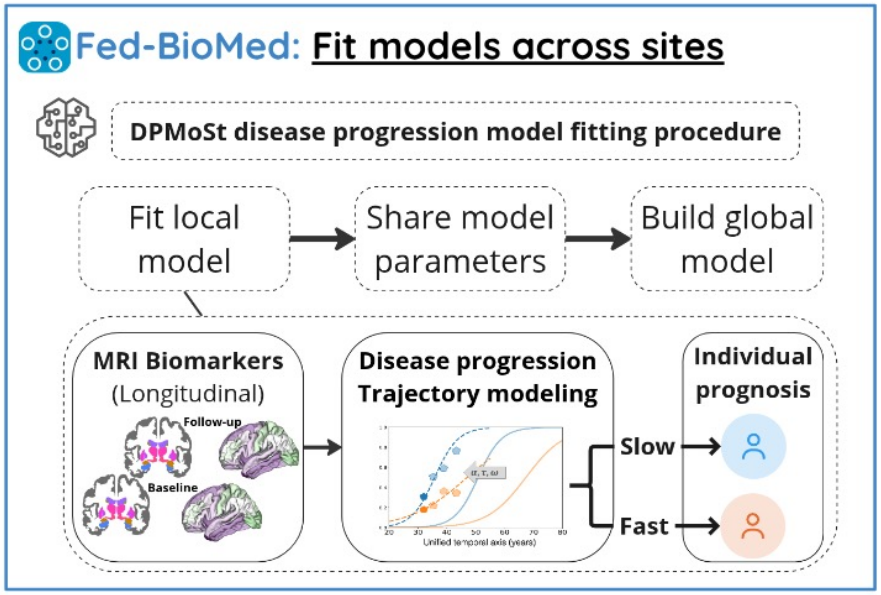
inherits Neurobagel vocabulary and semantics
standardizes local data
harmonizes data preprocessing

ORIGAMI

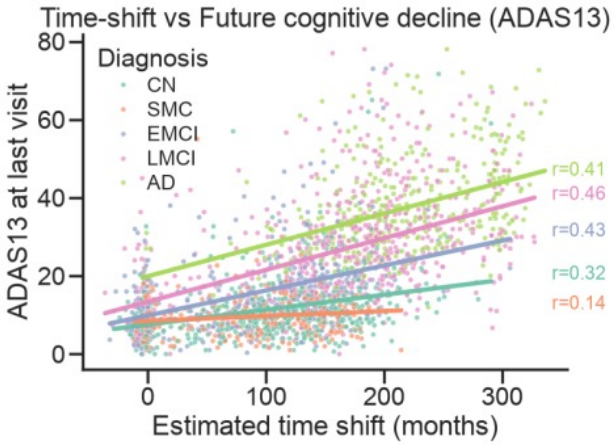
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Neuroimaging Application



$$b_{i,j}(t) = \frac{1}{1 + e^{-k_j(t+\tau_i-m_j)}} + \epsilon$$



Nipoppy towards Fed-BioMed

- I/O Python API
- Expose meta-data for Fed-BioMed query

Fed-BioMed towards Nipoppy

- Federated Data Loader
- Federated Data Query

Wang et al. OHBM 2026

Federated Exploration and Analytics

```
exp.training_data().data()
```

```
{'NODE_e1d11980-12dd-4fde-a356-6554d68c593d': {'name': 'csv',
  'data_type': 'csv',
  'tags': ['tabular'],
  'description': 'tabular dataset',
  'shape': {'csv': [10668, 7]},
  'dtypes': {'year': 'Int64',
   'price': 'Int64',
   'transmission': 'Int64',
   'mileage': 'Int64',
   'tax': 'Int64',
   'mpg': 'Float64',
   'engineSize': 'Float64'},
  'dataset_id': 'dataset_0ca1dd3b-6bb5-41a4-86ef-2f6b3a905ff3',
  'dataset_parameters': {}},
 'NODE_d0e82145-2311-46cb-93f8-922f36f4b71d': {'name': 'cars',
  'data_type': 'csv',
  'tags': ['tabular'],
```

```
# Which stat keys are stored at leaves (across all nodes)?
print('Available stats:', result.available_stats())

# Which can be aggregated globally?
print('Computable stats:', result.computable_stats())
```

```
Available stats: ['count', 'mean', 'variance']
Computable stats: ['count', 'mean', 'std', 'sum', 'variance']
```

```
# Raw per-node outputs — useful for debugging or site-level comparisons
result.node_stats()
```

```
{'NODE_e1d11980-12dd-4fde-a356-6554d68c593d': {'year': {'mean': 2017.100684353619,
   'count': 10667,
   'variance': 4.6984687},
```

The collaborative learning pipeline

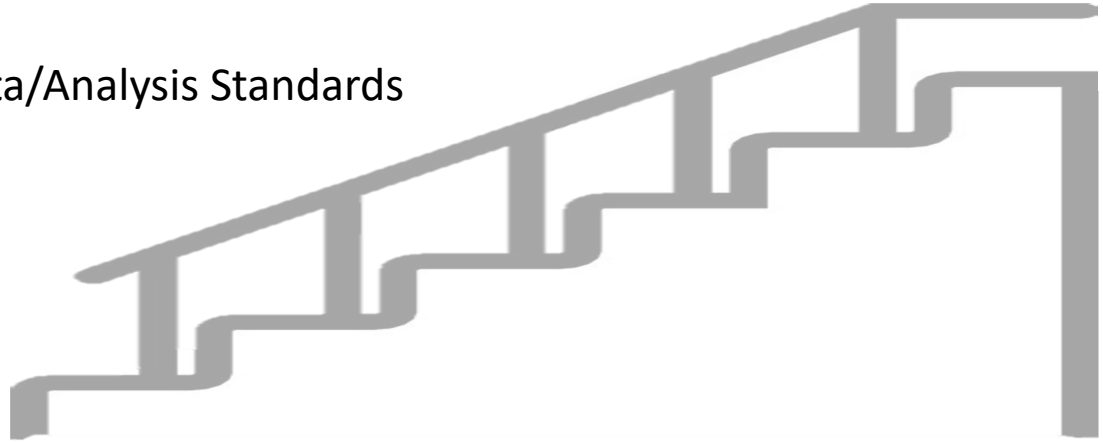
Cost effectiveness



Harmonization

Data/Analysis Standards

Data Access

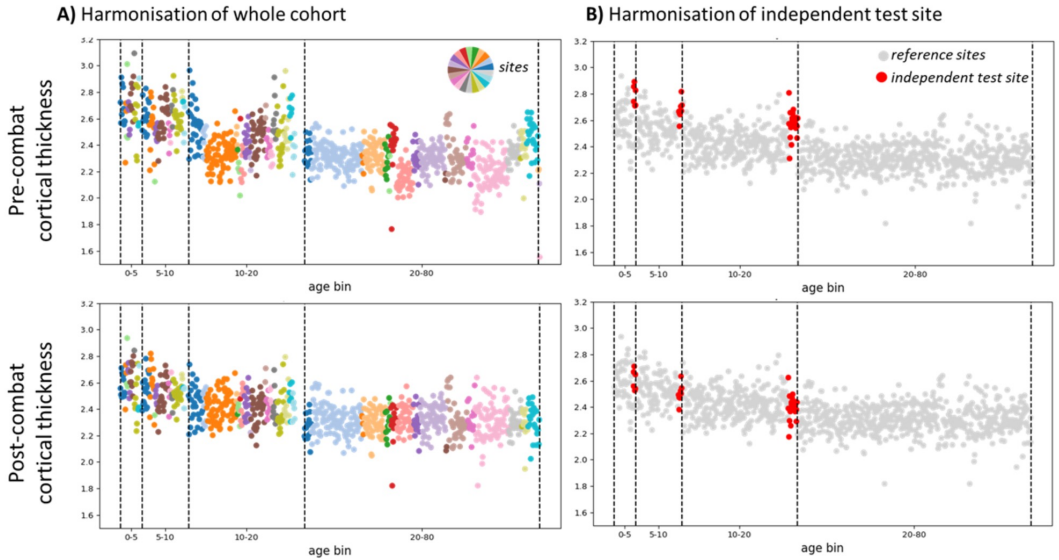


From Federated Data Harmonization

ComBat

$$y_{ijg} = \alpha_g + \phi(\mathbf{x}_{ij}, \boldsymbol{\theta}_g) + \gamma_{ig} + \delta_{ig}\varepsilon_{ijg}$$

$$y_{ijg}^{\text{ComBat}} = \frac{\hat{\sigma}_g}{\hat{\delta}_{ig}^*}\left(z_{ijg} - \hat{\gamma}_{ig}^*\right) + \hat{\alpha}_g + \phi(\mathbf{x}_{ij}; \hat{\boldsymbol{\theta}}_g)$$



Spitzer et al. *Brain*, 2022
Fortin et al. *Neuroimage*, 2019
Johnson et al. *Biostatistics*, 2007

Silva S., et al. DOI:10.1101/2023.05.24.542107

From Federated Data Harmonization

UCL

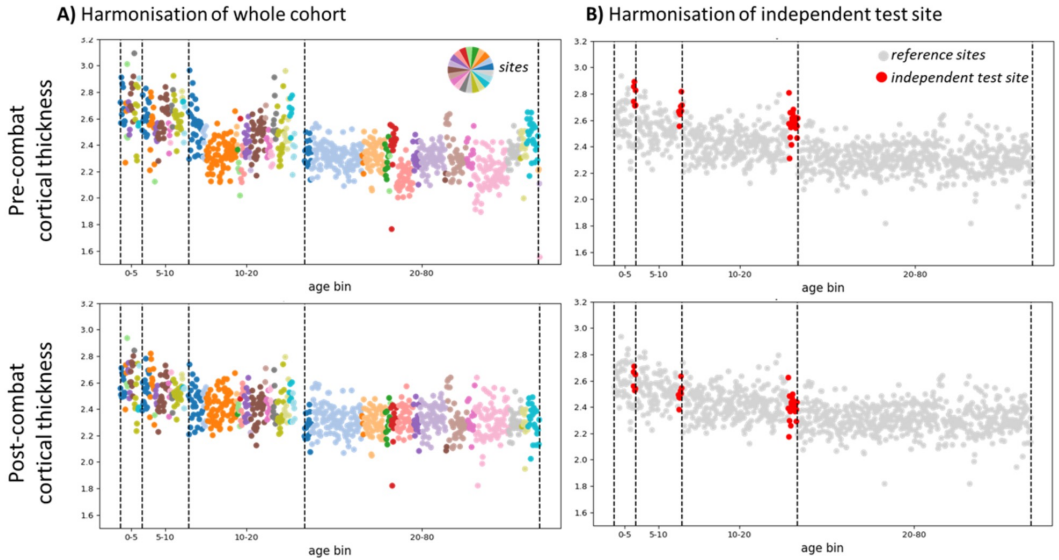
Federated ComBat

$$y_{ijg} = \alpha_g + \phi(\mathbf{x}_{ij}, \boldsymbol{\theta}_g) + \gamma_{ig} + \delta_{ig}\varepsilon_{ijg}$$

$$y_{ijg}^{\text{ComBat}} = \frac{\hat{\sigma}_g}{\hat{\delta}_{ig}^*}\left(z_{ijg} - \hat{\gamma}_{ig}^*\right) + \hat{\alpha}_g + \phi(\mathbf{x}_{ij}; \hat{\boldsymbol{\theta}}_g)$$

local processing

federated optimization



Spitzer et al. *Brain*, 2022
Fortin et al. *Neuroimage*, 2019
Johnson et al. *Biostatistics*, 2007

Silva S., et al. DOI:10.1101/2023.05.24.542107

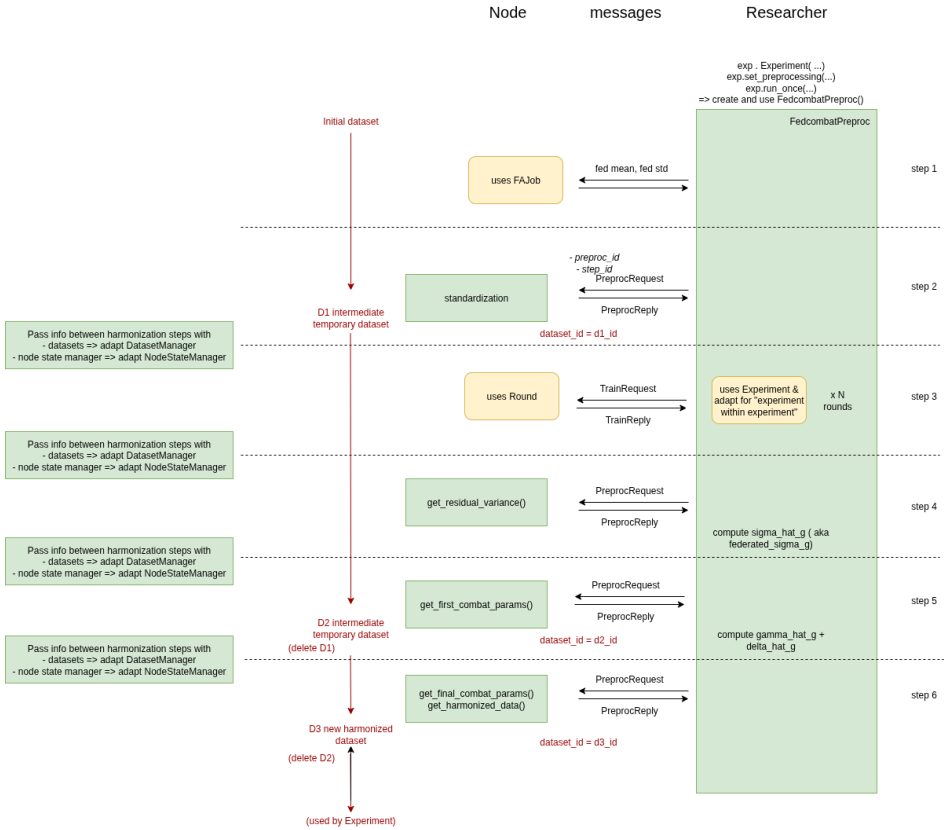
From Federated Data Harmonization

Federated ComBat

$$y_{ijg} = \alpha_g + \phi(\mathbf{x}_{ij}, \boldsymbol{\theta}_g) + \gamma_{ig} + \delta_{ig}\varepsilon_{ijg}$$

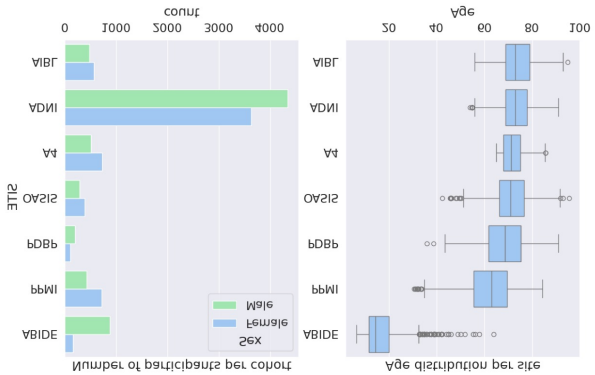
$$y_{ijg}^{\text{ComBat}} = \frac{\hat{\sigma}_g}{\hat{\delta}_{ig}^*}\left(z_{ijg} - \hat{\gamma}_{ig}^*\right) + \hat{\alpha}_g + \phi(\mathbf{x}_{ij}; \hat{\boldsymbol{\theta}}_g)$$

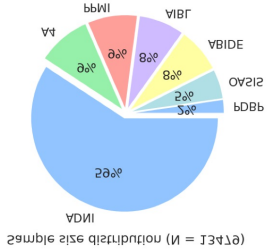
local processing — federated optimization



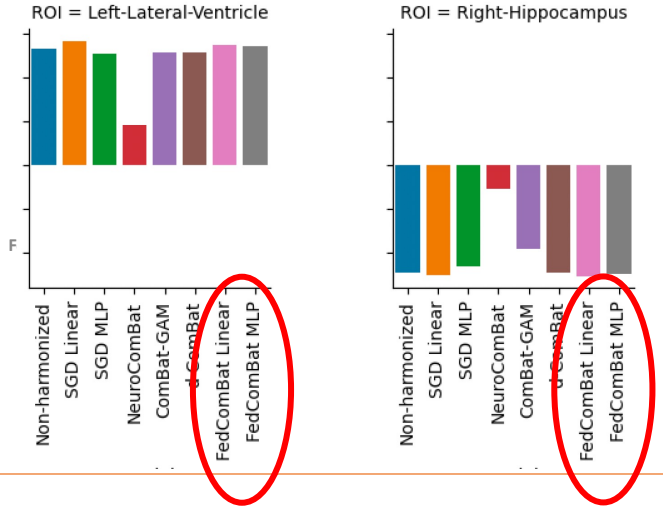
Silva S., et al. DOI:10.1101/2023.05.24.542107

From Federated Data Harmonization

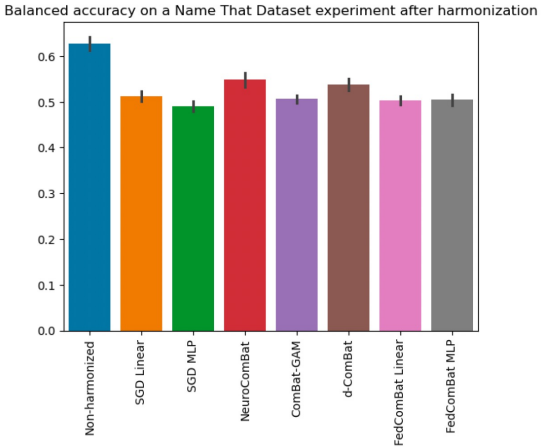




Preserving meaningful relations…



…while removing site effects



The collaborative learning pipeline

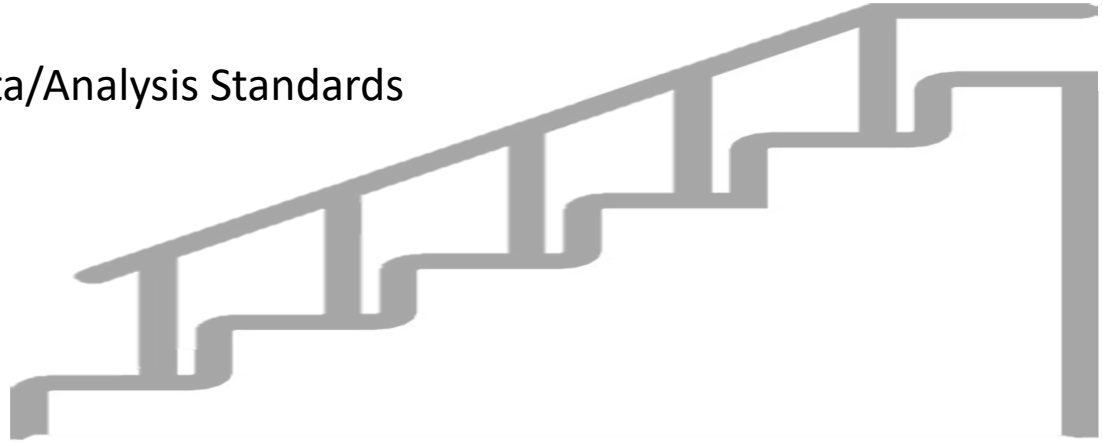


Cost effectiveness

Harmonization

Data/Analysis Standards

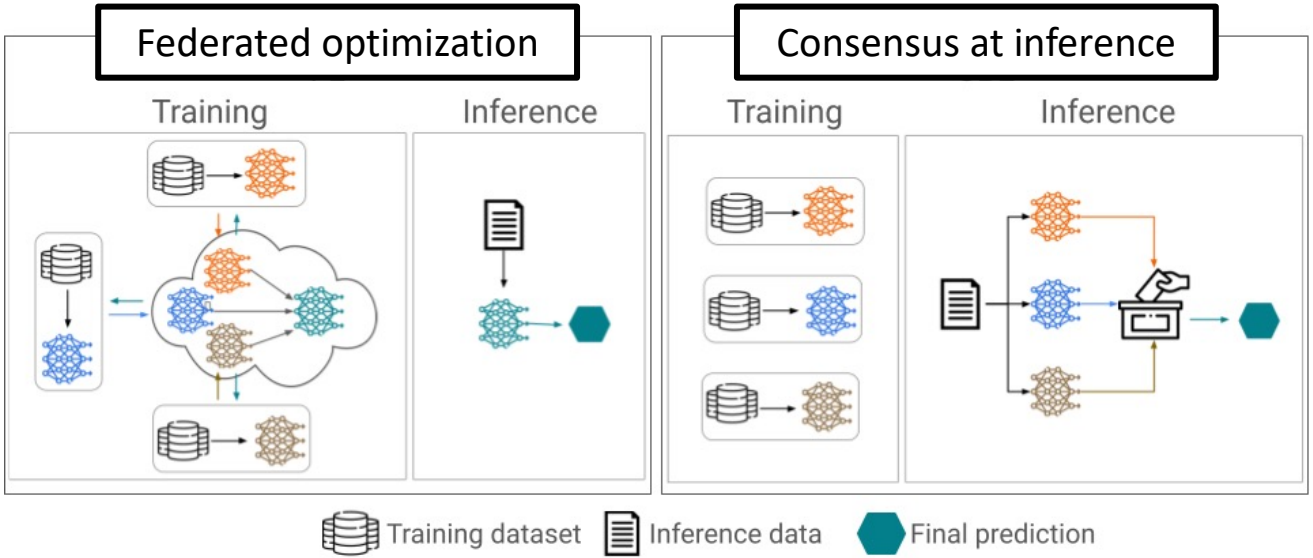
Data Access



[Submitted on 9 Dec 2024 (v1), last revised 8 Sep 2026 (this version, v3)]

A cautionary tale on the cost-effectiveness of collaborative AI in real-world medical applications

Francesco Cremonesi, Lucia Innocenti, Sebastien Ourselin, Vicky Goh, Michela Antonelli, Marco Lorenzi



Computer Science > Machine Learning

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npj Scientific Reports, 2026

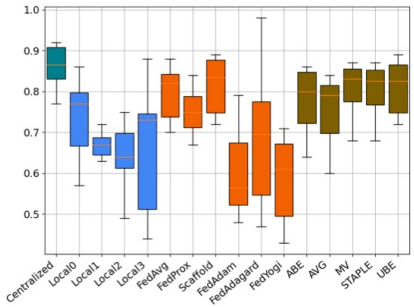
Dataset	#Clients	Dataset size per client	Data Modality	Task *[Metric]*
FedProstate [15, 16, 17]	6	[32, 23, 27, 184, 5, 36]	T2 MRI	Segmentation *[DiceScore]*
FedHeart [18]	4	[303, 261, 46, 130]	Tabular Data	Classification *[Accuracy]*
FedIXI [18]	3	[311, 181, 74]	T1 MRI	Segmentation *[DiceScore]*
FedISIC [18]	6	[12k, 3.9k, 3.3k, 225, 819, 439]	Dermoscopy images	Classification *[Accuracy]*
FedTCGA-BRCA [18]	6	[311, 196, 206, 162, 162, 51]	Tabular Data	Survival *[C-Index]*
FedKiTS [18]	6	[12, 14, 12, 12, 16, 30]	CT Scans	Segmentation *[DiceScore]*
FeTS [19]	23	4 to 511 *(see Suppl.)*	T1, T1CE, T2 & FLAIR MRIs	Segmentation *[DiceScore]*

FLamby

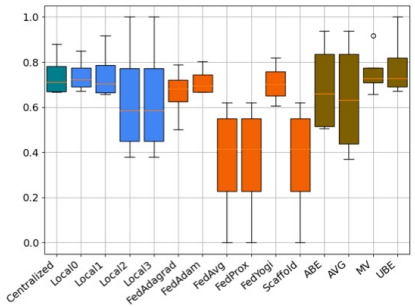
https://github.com/owkin/FLamby

Method	CL Paradigm	Reference
FedAvg	FL	McMahan et. Al., 2017 [20]
FedProx	FL	Li et Al., 2018 [21]
scaffold	FL	Karimireddy et Al., 2020 [22]
FedAdam	FL	Reddi et Al., 2020 [23]
FedAdagrad	FL	Reddi et Al., 2020 [23]
FedYogi	FL	Reddi et Al., 2020 [23]
Avg	CBL	Guha et Al., 2019 [7]
Mv	CBL	Safdar et Al., 2021 [24]
Staple	CBL	Warfield et Al., 2004 [25]
Ube	CBL	*Inspired by* Ruta et Gabrys, 2000 [26]
Abe	CBL	*Inspired by* Ruta et Gabrys, 2000 [26]

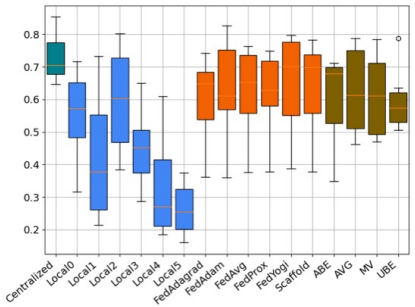
Federated Training vs Consensus Inference



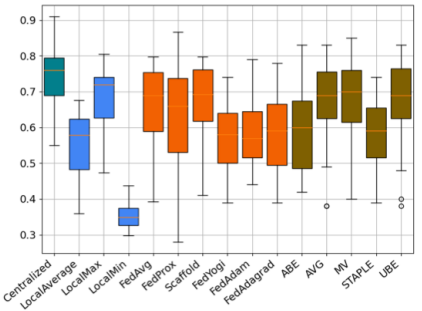
(a) FedProstate



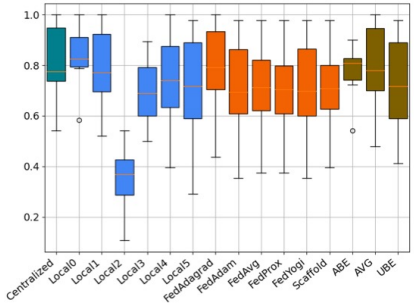
(b) FedHeart



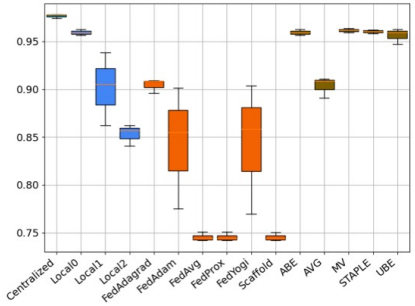
(c) FedIsic



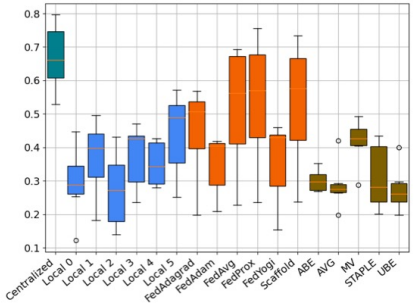
(g) FeTS



(d) FedTGCA-BRCA



(e) FedIXI



(f) FedKiTS

Federated Training vs Consensus Inference

Experiment	Average performance ratio ($\frac{\text{CBL}}{\text{FL}}$)	95% CI	U statistic	p-value
FedProstate	1.12×	[1.015, 1.245]	22.5	.20
FedHeart	1.36×	[1.070, 1.801]	19	.16
FedIXI	1.18×	[1.105, 1.258]	29	.01
FedISIC	0.97×	[0.950, 1.001]	5	.17
FedTCGA-BRCA	1.06×	[1.003, 1.112]	10	.10
FedKITS	0.70×	[0.581, 0.855]	2	.02
FeTS	1.03×	[0.957, 1.107]	19	.54

Task	Average performance ratio $\frac{\text{CBL}}{\text{FL}}$	95% CI	U statistic	p-value
CLASSIFICATION	1.11	[1.014, 1.228]	18	.26
SEGMENTATION	1.02	[0.956, 1.099]	17	.79
SURVIVAL	1.06	[1.003, 1.113]	16	.10

Federated Training vs Consensus Inference

Experiment	Paradigm	Best method	Metric	95% CI	Common method	Metric	95% CI
FedProstate	CBL	UBE	0.81	[0.758, 0.860]	AVG	0.75	[0.680, 0.817]
	FL	Scaffold	0.82	[0.760, 0.868]	FedAVG	0.80	[0.743, 0.850]
FedHeart	CBL	UBE	0.78	[0.684, 0.924]	AVG	0.64	[0.416, 0.869]
	FL	FedAdam	0.72	[0.667, 0.769]	FedAVG	0.36	[0.132, 0.574]
FedIXI	CBL	MV	0.96	[0.959, 0.964]	AVG	0.90	[0.891, 0.911]
	FL	FedAdagrad	0.90	[0.896, 0.909]	FedAVG	0.75	[0.742, 0.751]
FedISIC	CBL	AVG	0.62	[0.522, 0.729]	AVG	0.62	[0.525, 0.728]
	FL	FedYogi	0.65	[0.518, 0.761]	FedAVG	0.62	[0.512, 0.726]
FedTCGA-BRCA	CBL	AVG	0.79	[0.643, 0.927]	AVG	0.79	[0.633, 0.927]
	FL	FedAdagrad	0.78	[0.619, 0.921]	FedAVG	0.70	[0.553, 0.846]
FedKITS	CBL	MV	0.42	[0.362, 0.461]	AVG	0.29	[0.239, 0.348]
	FL	FedProx	0.54	[0.384, 0.672]	FedAVG	0.52	[0.375, 0.648]
FeTS	CBL	MV	0.67	[0.617, 0.715]	AVG	0.66	[0.612, 0.711]
	FL	Scaffold	0.67	[0.630, 0.715]	FedAVG	0.66	[0.609, 0.701]

Federated Training vs Consensus Inference

	Training Time [min]				**Bandwidth** [MB]		
	FL	CBL	FL Slow factor	95% CI	FL	CBL	FL Increment
FedProstate	1.2×10^3	1.1×10^2	8×	[4.6, 15.6]	4.0×10^3	1.15×10^2	35×
FedHeart	0.15	0.017	9×	[5.8, 15.7]	13.1	0.09	146×
FedIXI	5.5	0.65	8×	[5.2, 22.0]	4.5×10^2	3	150×
FedISIC	193	10.2	19×	[10.0, 50.3]	4.06×10^4	1.18×10^2	345×
FedTCGA-BRCA	0.455	0.017	26×	[21.2, 34.8]	5.42	0.15	36×
FedKITS	1581	54.1	29×	[21.9, 37.2]	8.8×10^4	7.4×10^2	120×
FeTS	3.4×10^4	2.9×10^3	12×	[10.0, 15.1]	3.7×10^4	4.5×10^2	83×

The collaborative learning pipeline

Data Access	Transparent, auditable rules across frameworks	Depends on trusted infrastructure for the policy engine (ledgers, secure hardware)
Data/Analysis Standards	Shared pipelines and metadata	Requires every site to adopt common processing
Harmonization	Generalizes linear and non-linear ComBat with no need to centralize data.	Challenge of extreme site heterogeneity
Cost Effectiveness	Comparable accuracy of Consensus at about 15× lower training time and 60× lower bandwidth.	Need for privacy guarantees for consensus. No paradigm wins everywhere



It takes a community to learn from data

Progress depends on shared standards more than on better algorithms

Research projects involving Fed-BioMed

Funded by the European Union

EUCAIM CANCER IMAGE EUROPE

Horizon EU Project: “EUropean federation of CAncer IMages” Oncology infrastructure with **66 EU research and clinical institutions.**

PI: Luis Marti-BONMANTI (Hospital Universitario y Politecnico La Fe de Valancia. Spain),

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anr agence nationale de la recherche

FRANCE 2030 PROGRAMME DE RECHERCHE SANTÉ NUMÉRIQUE

PEPR e-Health research program - Secure, Safe and fair machine learning for healthcare

8 partners including CEA, CNRS, Inria. PI: Olivier Cappé (CNRS)

AMGEN INNOVATIONS LA SCIENCE & L'HUMAIN

unicancer

PRED-IRM - Prediction of non-responders to glioblastoma treatment using pre-treatment IMR data with **5 centers in France**

PI: Aurelien Corroyer-Dulmont, Centre François Baclesse

---

anr agence nationale de la recherche

ANR project I-Vesseg, Vascular brain analysis across **4 centers in France, UK and Italy.** *EURECOM, Grenoble Institute of Neurology, UCL, Siena University, PI: M.A. Zuluaga (EURECOM, France)*

DTRIP4H – EU Horizon Project

Research projects involving Fed-BioMed

Decentralised Digital Twin in existing Research Infrastructures for Predictive, Preventive, Personalised, and Participatory Health.





DTRIP4H
DIGITAL TWINS FOR BETTER HEALTH

Funded by
the European Union

Horizon EU Project grouping **23 academic research, clinical and industry partners**
PI: Dr. Pirjo Spuul (Tallinna Tehnikaülikool.)

Why contribute to Fed-BioMed ?

Shaping the Future of Medicine with Collaborative Learning

Be part of an initiative **tackling societal needs** in healthcare

Support an open-source, transparent, community-driven option for the healthcare domain

Increase the visibility and impact of your research

Fuel for thoughts with new scientific and technical challenges for your work

Collaborate at the crossroads of academia, healthcare, industry, and open-source

Custom collaboration scheme

Different paths to join the Fed-BioMed community



Use
provide feedback

Contribute code
bugfix,
functionality

Setup common project
long term, financed

Join governance
global engagement in
project

Thank you!

fedbiomed.org

github.com/fedbiomed

```
pip install fedbiomed[node, researcher]
```



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FRANCE 2030 PROGRAMME DE RECHERCHE SANTÉ NUMÉRIQUE